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基于 WebGL 的三维地质建模及可视化方法研究

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摘 要

随着信息技术的迅猛发展和数字矿山建设的深入推进，三维地质建模技术已经成为地质勘探、矿产资源评价、矿山工程设计、安全生产管理以及灾害预测与防治等关键环节不可或缺的技术支撑。该技术能够将分散、抽象的地质勘探数据转化为直观、连续的三维空间模型，为深刻理解地下复杂的地质结构、精确揭示矿产资源的赋存规律、科学优化矿产开采方案提供了强大的可视化分析手段与决策依据。当前，众多矿山企业已普遍部署并应用了基于 Web 架构的监测监控系统、地质信息保障平台、生产调度指挥中心以及数据集成中控平台等，初步实现了矿山各类信息的集中化管理和便捷的远程访问。然而，在面向三维地质建模与可视化这一专业领域时，现有解决方案普遍存在显著的局限性。传统的桌面端专业软件虽然功能强大，但往往存在平台依赖性强、软件授权成本高、部署与维护过程复杂、数据共享与协同工作不便等问题。这些因素共同制约了三维地质模型价值的充分发挥，特别是在需要跨部门协作、异地远程访问、轻量化部署以及与现有 Web 信息系统无缝集成的应用场景下，迫切需要一种更为先进和适应性更强的技术方案。WebGL（Web Graphics Library）技术的出现及其生态的日益成熟，为突破上述瓶颈提供了革命性的契机。作为一种开放的、免插件的 Web 图形标准，WebGL 允许在兼容的网页浏览器内直接利用计算机的图形处理器（GPU）进行硬件加速的二维和三维图形渲染，结合成熟的浏览器/服务器（B/S）架构，能够构建出跨平台、跨设备、高性能、易部署、易集成的 Web 端三维可视化应用。因此，深入研究基于 WebGL 的三维地质建模核心方法及其高效可视化技术，并构建适应 Web 环境的交互式三维地质模型平台，对于提升数字矿山的信息化与智能化水平、促进地质数据的有效共享与深度应用、辅助矿山进行科学决策具有重要的理论研究价值和迫切的现实应用意义。

国内外在地质建模与可视化领域的研究已取得长足进步。建模方法从早期的规则网格插值，发展到基于不规则三角网（TIN）、边界表示（B-Rep）、体元（Voxel）乃至混合表示等多种技术路径，其中基于 TIN 的方法因其对不规则散乱数据的良好适应性和对地质体表面形态的精确表达而得到广泛应用，约束 Delaunay 三角剖分（CDT）作为其中的关键算法，能够有效处理断层等地质构造对地层形态的约束。可视化方面，传统桌面软件功能强大但灵活性不足，而 WebGL 的兴起为 Web 端三维地质应用带来了新的可能性，但针对地质领域特有的复杂模型（如大规模地层、复杂相交断层网络、富含信息的钻孔柱状图、地下巷道系统等）的精细化建模、Web 端高效渲染、逼真纹理映射优化以及专业地质分析功能的集成，仍然面临诸多技术挑战，亟待深入研究和突破。

本研究致力于系统性地探索并实践一套完整的、面向 Web 环境的、基于 WebGL 的三维地质建模与可视化解决方案。研究工作覆盖了从原始地质数据的预处理、三维地质实体的精细化构建、Web 端的高效可视化渲染到交互式分析功能开发的全流程，并重点关注其中的关键技术难点与性能优化策略。首先，在数据层面，设计了能够容纳钻孔、断层、地层等多种地质信息的基础数据结构，并研究了钻孔数据的自动化、批量化处理方法，高效提取空间坐标、地层分界点、岩性属性等建模所需关键信息。在三维地质实体建模环节，针对不同地质要素的特点采用了差异化的建模策略：对于钻孔模型，提出了一种基于分段圆柱体的可视化建模方法，将钻孔轨迹的空间位置与沿深度方向的地层分层信息相结合，通过不同颜色或纹理的圆柱段直观展示钻孔穿过的地层及其属性；对于断层模型，重点研究了复杂断层系统，特别是断层之间空间相交关系的处理方法，通过对相交情况进行分类（如 T 型、X 型相交等），精确计算断层面之间的交线，并将这些交线作为强制约束边界，结合断层上下盘边界点的区分处理，利用约束 Delaunay 三角剖分算法生成能够准确反映断层几何形态及其对围岩切割关系的三角网格模型；对于地层模型，则基于处理后的钻孔地层分界点数据集以及由断层模型提供的约束边界数据，同样采用约束 Delaunay 三角剖分算法，生成各地层顶、底板的三角网格模型，构建出层状地质体；此外，还实现了对外部常用格式（如 OBJ）的巷道模型的导入与整合功能，使其能够无缝融入统一的三维地质场景中。

在可视化与交互实现方面，本研究选择 WebGL 作为底层的图形渲染 API，并基于应用广泛且生态成熟的 Three.js JavaScript 库进行二次开发与功能扩展，以更好地满足地质模型可视化的特定需求。充分利用 WebGL 提供的 GPU 硬件加速能力，实现大规模、复杂地质模型的高效渲染，确保用户在 Web 端获得流畅的视觉体验。实现了对地层（三角网格模型）、断层（三角网格模型）、钻孔（分段圆柱体模型）以及巷道（导入的外部模型）等多种地质要素的可视化表达，并研究了光照模型、材质属性（如颜色、透明度、光泽度等）的合理设置，以增强模型的真实感和信息传达的清晰度。针对地质模型（尤其是形态不规则的地层和断层面）纹理映射中 UV 坐标计算的难题，即如何自动生成适应性良好、减少纹理拉伸与畸变的 UV 坐标，本研究提出了一种创新性的结合模型包围盒（Bounding Box）信息与表面法向量（Normal Vector）信息的自适应 UV 坐标计算方法。该方法旨在通过几何分析自动确定最优的投影方式和参数，以期在无需大量人工干预的情况下，显著提升纹理贴图在地质模型表面的视觉效果与真实感。在交互功能方面，系统设计并实现了包括场景漫游（提供基于鼠标的轨道控制器用于全局浏览，以及基于第一人称视角的键盘/鼠标控制用于巷道内部的沉浸式漫游，并考虑了基本的碰撞检测以增强真实感）、模型基本操作（旋转、缩放、平移）、图层管理（对

不同类型的地质模型进行独立的显示/隐藏控制、透明度调节）、以及基于射线拾取（Raycasting）技术的交互式属性查询（用户可通过鼠标点击三维场景中的任意模型，系统即时反馈该位置对应的钻孔信息、地层属性或断层参数等详细信息）等一系列基础且必要的交互功能。

为了满足矿井水害防治等实际生产应用中对地层厚度或层间距信息的迫切需求，本研究特别设计并实现了一种基于射线与三角网格求交检测的地层层间距（或指定地层的厚度）的动态计算方法。该方法允许用户在三维可视化环境中通过交互方式（如鼠标点击）指定查询点或区域，系统能够自动沿预设方向（通常是垂直方向，或根据地层产状沿法线方向）发射虚拟射线，精确计算该射线与目标地层顶、底板（或上、下两个相邻地层）的三角网格模型的交点，进而得到该位置的准确层间距或厚度值。计算结果不仅可以以数值形式反馈给用户，还可以通过颜色映射（Color Mapping）等可视化手段，直观地渲染在模型表面或生成专题图，清晰展示地层厚度或层间距在空间上的分布规律与变化趋势，为识别潜在的薄弱区域、进行工程稳定性分析和水害风险评估提供定量的、可视化的数据支撑。

在系统实现层面，本研究采用成熟的浏览器/服务器（B/S）架构，设计了前后端分离的系统框架，提高了系统的灵活性和可维护性。前端利用现代 Web 技术栈（HTML5, CSS3, JavaScript）以及核心的 Three.js 库构建用户交互界面和三维渲染引擎。在系统开发过程中，高度关注 Web 端的运行性能，综合运用了多种前端性能优化技术来应对大规模地质数据带来的挑战。具体措施包括：采用异步数据加载机制，避免在加载模型和纹理等大资源时阻塞用户界面的响应；实施有效的场景管理策略，例如根据视距和模型重要性进行动态渲染（尽管原文未明确提及，但类似 LOD 的思想是常见的优化方向）；优化 Three.js 场景图管理，特别是注重对不再使用的几何体、材质、纹理等 GPU 资源的及时检测与显式释放，以减少内存泄漏风险和 GPU 显存占用，降低渲染负载；探索模型数据压缩传输等手段减少网络传输时间和带宽消耗。通过这些优化措施，旨在确保即使在处理包含数百万甚至更多三角面片的复杂地质模型时，系统也能在主流配置的计算机和现代浏览器上保持流畅的运行速度和良好的交互响应性。

为了全面验证本研究所提出的理论方法、关键技术以及开发的系统平台的有效性和实用性，选取了中国河北省邢台市某矿区的实际地质勘探数据作为研究背景和应用实例。利用该矿区详细的钻孔数据（包含坐标、高程、地层分层信息、岩性描述等）、已探明的断层构造数据以及规划或已建成的巷道工程数据，成功构建了该区域内包含复杂断层网络结构、多套地层分布以及地下巷道系统的三维地质模型。随后，在开发的基于 WebGL 的原型系统中，对该模型进行了高质量的可视化展示，并提供了前述的各种交互式分析功能。特别地，对核心的地层层间

距计算功能进行了重点验证。通过在模型上选取多个代表性位置进行层间距计算，并将计算结果与矿区内部分实际钻孔揭露的地层厚度数据进行了详细的对比分析。对比结果显示，本系统采用的基于射线求交方法计算得到的地层层间距与钻孔实测数据之间具有良好的一致性，误差在可接受范围内，充分证明了该计算方法的准确性和可靠性。整个应用实例的成功实施，不仅展示了本研究成果在处理真实世界复杂地质数据、构建高精度三维地质模型、提供直观交互式可视化方面的能力，也验证了所集成地质分析功能（如层间距计算）的实用价值。这表明，本研究提出的基于 WebGL 的 Web 端三维地质建模与可视化方案是可行的，并且能够为矿产资源的勘探评价、储量估算、矿山工程设计、开采方案优化、矿井水文地质分析及水害风险评估等实际工作提供重要的三维空间信息支持和强大的辅助决策工具。

研究结果表明，基于 WebGL 和 Three.js 的技术路线能够有效支撑 Web 端三维地质应用的开发，实现了地质模型的高效渲染与流畅交互。通过约束 Delaunay 三角剖分方法，并结合对断层相交关系的精细化处理，成功构建了能够反映复杂地质构造的断层和地层模型；基于分段圆柱体的钻孔模型直观展示了钻孔内部信息；提出的结合包围盒与法向量的自适应 UV 坐标计算方法，在一定程度上改善了地质模型的纹理贴图效果；实现的巷道漫游、属性查询、特别是基于射线求交的地层层间距计算等交互与分析功能，提升了系统的实用价值。同时，通过采用 GPU 加速渲染、异步加载、内存优化管理等一系列性能优化策略，有效解决了大规模地质数据在 Web 端加载与渲染的性能瓶颈问题，确保了系统的可用性和用户体验。邢台某矿的应用实例验证了本研究方法和系统的可行性、准确性与稳定性。

尽管本研究在 Web 端三维地质建模与可视化方面取得了显著进展，但仍存在若干值得进一步探索和完善的方向。首先，在模型构建精度方面，当前主要依赖的三角网格（TIN）结构在表达极其复杂的地质构造（如剧烈褶皱、大规模逆冲推覆构造、密集断裂带及其复杂交切关系）时，仍可能面临网格质量难以保证、拓扑关系维护困难等挑战。未来研究可考虑引入体元（Voxel）、隐式曲面（Implicit Surface）或混合建模（Hybrid Modeling）等更先进的表示方法，并探索如各向异性网格（Anisotropic Meshing）、自适应网格细化（Adaptive Mesh Refinement）、多分辨率模型（Multi-resolution Models）等高级网格技术，以期在保证模型精度的同时，优化计算开销和存储需求。其次，在应对超大规模地质数据集的实时交互与渲染方面，虽然现有的 GPU 加速和优化技术已显著提升性能，但面对未来可能出现的亿级甚至更高量级的地质模型数据，当前的 Web 端渲染能力可能仍会遇到瓶颈。未来可以密切关注并积极探索 WebGPU 等下一代 Web 图形 API，利用其提供的更底层硬件访问能力、更佳的多线程支持以及现代图形特性，进一步挖掘

Web 端的渲染潜力。同时，需要持续研究和优化数据流式加载（Streaming）、视锥剔除（View Frustum Culling）、遮挡剔除（Occlusion Culling）、精细化的 LOD（Level of Detail）策略以及服务器端渲染（Server-Side Rendering）等技术组合，以应对不断增长的数据规模挑战。再次，在纹理映射方面，本研究提出的自适应 UV 计算方法虽然有所改进，但在处理具有复杂曲率变化或存在拓扑奇异点（如尖锐边缘、顶点）的几何体时，仍可能产生投影失真或 UV 坐标不连续（跳变）的问题。未来的研究可以探索更先进、更鲁棒的曲面参数化算法（如共形映射、最小化面积或角度畸变的方法），或者尝试结合机器学习技术来自动优化 UV 布局。同时，也需要权衡算法复杂性、计算效率与最终视觉效果，可能需要在预处理阶段进行更精细的模型分割或特征识别来辅助 UV 生成。此外，随着 Web 技术的持续演进，基于 WebGL（及未来的 WebGPU）的三维地质建模平台有望进一步扩展其功能边界和应用场景。例如，可以深度融合虚拟现实（VR）与增强现实（AR）技术，为用户提供更加沉浸式、交互性更强的地质模型浏览、分析和协同工作体验，使得地质专家能够“身临其境”地探索地下世界。还可以加强与人工智能（AI）技术的结合，探索地质模型的智能解译、地质异常的自动识别、基于地质知识的智能分析与预测等高级应用。最后，应着力推动该类平台与矿山现有的其他 Web 信息系统（如监测监控、生产管理、安全预警等）实现更深层次的数据集成与业务联动，通过将三维地质模型与实时生产数据、传感器监测数据等进行融合可视化与关联分析，构建真正意义上的矿山数字孪生（Digital Twin）体，从而为矿山的智能化运营和精细化管理提供更为全面、动态、智能的支撑。

关键词：WebGL；三维地质建模；可视化；纹理映射；巷道漫游

Abstract

With the rapid development of information technology and the continuous advancement of digital mine construction, 3D geological modeling technology has become an indispensable core technical support in crucial areas such as geological exploration, mineral resource assessment, mine engineering design, safe production management, and disaster prediction and prevention. This technology transforms discrete, abstract geological exploration data into intuitive, continuous 3D spatial models, providing powerful visualization and analysis tools for profoundly understanding complex subsurface geological structures, accurately revealing the occurrence patterns of mineral resources, and scientifically optimizing mining plans. Currently, numerous mining enterprises have widely deployed and utilized Web-based systems for monitoring and control, geological information assurance platforms, production dispatch command centers, and integrated data control platforms, initially achieving centralized management of various mine information and convenient remote access. However, when addressing the specialized field of 3D geological modeling and visualization, existing solutions commonly exhibit significant limitations. While powerful, traditional desktop professional software often suffers from strong platform dependency, high software licensing costs, complex deployment and maintenance processes, and difficulties in data sharing and collaborative work. These factors collectively restrict the full realization of the value of 3D geological models, particularly in application scenarios requiring inter-departmental collaboration, remote access from different locations, lightweight deployment, and seamless integration with existing Web information systems, thus urgently necessitating a more advanced and adaptable technical solution. The emergence of WebGL (Web Graphics Library) technology and its increasingly mature ecosystem offers a revolutionary opportunity to overcome these bottlenecks. As an open, plugin-free Web graphics standard, WebGL allows for hardware-accelerated 2D and 3D graphics rendering directly within compatible web browsers using the computer's Graphics Processing Unit (GPU). Combined with the mature Browser/Server (B/S) architecture, it enables the construction of cross-platform, cross-device, high-performance, easily deployable, and easily integrable Web-based 3D visualization applications. Therefore, in-depth research into core WebGL-based 3D geological modeling methods and their efficient visualization techniques, along with the construction of an interactive 3D geological model platform adapted to the Web environment, holds significant theoretical research value and pressing practical significance for enhancing the informatization and intelligence levels of digital mines, promoting the effective sharing and deep application of geological data, and supporting scientific decision-making in mining operations.

Research in geological modeling and visualization has made considerable progress both domestically and internationally. Modeling methods have evolved from early regular grid interpolation to various technical paths based on Triangulated Irregular Networks (TIN), Boundary Representation (B-Rep), Voxel, and even hybrid representations. Among these, TIN-based methods are widely used due to their good adaptability to irregularly scattered data and accurate representation of geological body surface morphology. Constrained Delaunay Triangulation (CDT) is a key algorithm within this approach, effectively handling constraints imposed by geological structures like faults on stratum morphology. In terms of visualization, traditional desktop software is powerful but lacks flexibility, while the rise of WebGL has brought new possibilities for Web-based 3D geological applications. However, significant technical challenges remain in the fine-grained modeling of complex geological models specific to the field (such as large-scale strata, complex intersecting fault networks, information-rich borehole logs, underground roadway systems), efficient rendering on the Web end, optimization of realistic texture mapping, and the integration of professional geological analysis functions, all requiring further in-depth research and breakthroughs.

This research is dedicated to systematically exploring and implementing a complete, Web-oriented, WebGL-based solution for 3D geological modeling and visualization. The research scope covers the entire process from preprocessing raw geological data, finely constructing 3D geological entities, efficiently rendering visualizations on the Web end, to developing interactive analysis functions, with a particular focus on key technical difficulties and performance optimization strategies. Firstly, at the data level, a fundamental data structure capable of accommodating various geological information such as boreholes, faults, and strata was designed. Methods for automated and batch processing of borehole data were investigated to efficiently extract key modeling information like spatial coordinates, stratum boundary points, and lithological attributes. In the 3D geological entity modeling phase, differentiated modeling strategies were adopted based on the characteristics of different geological elements: for borehole models, a visualization modeling method based on segmented cylinders was proposed, combining the spatial trajectory of the borehole with stratum layering information along the depth, intuitively displaying the strata penetrated by the borehole and their attributes using cylinders of different colors or textures; for fault models, the focus was on complex fault systems, especially the handling of spatial intersection relationships between faults. By classifying intersection scenarios (e.g., T-type, X-type intersections), the intersection lines between fault planes were accurately calculated. These lines, along with differentiated processing of hanging wall and footwall boundary points, were used as mandatory constraint edges in the Constrained Delaunay Triangulation algorithm to generate triangular mesh models that accurately reflect the fault's geometry and its cutting relationship with the surrounding rock; for stratum models, based on the processed borehole stratum boundary point dataset and the constraint boundary data provided by the fault models, the Constrained Delaunay Triangulation algorithm was also employed to generate triangular mesh models for the top and bottom surfaces of each stratum, constructing layered geological bodies; additionally, the functionality to import and integrate external roadway models in common formats (like OBJ) was implemented, allowing them to be seamlessly incorporated into the unified 3D geological scene.

In terms of visualization and interaction implementation, this study selected WebGL as the underlying graphics rendering API and conducted secondary development and functional extension based on the widely used and mature Three.js JavaScript library to better meet the specific needs of geological model visualization. Leveraging the GPU hardware acceleration capabilities provided by WebGL, efficient rendering of large-scale, complex geological models was achieved, ensuring a smooth visual experience for users on the Web end. Visualization representations were implemented for various geological elements, including strata (triangular mesh models), faults (triangular mesh models), boreholes (segmented cylinder models), and roadways (imported external models). Reasonable settings for lighting models and material properties (such as color, transparency, shininess) were investigated to enhance the realism of the models and the clarity of information conveyed. Addressing the challenge of UV coordinate calculation for texture mapping on geological models (especially irregularly shaped strata and fault surfaces) – specifically, how to automatically generate UV coordinates that are well-adapted and minimize texture stretching and distortion – this research proposed an innovative adaptive UV coordinate calculation method combining model Bounding Box information with surface Normal Vector information. This method aims to automatically determine the optimal projection method and parameters through geometric analysis, significantly improving the visual effect and realism of texture maps on geological model surfaces without requiring extensive manual intervention. Regarding interactive functionalities, the system was designed and implemented with a series of fundamental and necessary interactive features, including scene navigation (providing a mouse-based orbit controller for global browsing, and keyboard/mouse control based on a first-person perspective for immersive roaming inside roadways, considering basic collision detection to enhance realism), basic model operations (rotation, scaling, translation), layer management (independent display/hiding control and transparency adjustment for different types of geological models), and interactive attribute querying based on ray casting technology (users can click on any model in the 3D scene with the mouse, and the system instantly provides detailed information corresponding to that location, such as borehole data, stratum properties, or fault parameters).

To meet the urgent need for information on stratum thickness or interlayer distance in practical production applications like mine water hazard prevention and control, this study specifically designed and implemented a dynamic calculation method for interlayer distance (or the thickness of a specified stratum) based on ray intersection detection with triangular meshes. This method allows users to specify query points or areas within the 3D visualization environment through interaction (e.g., mouse clicks). The system can automatically cast virtual rays along a predefined direction (typically vertical, or along the normal direction based on stratum occurrence) and accurately calculate the intersection points of these rays with the triangular mesh models of the target stratum's top and bottom surfaces (or two adjacent strata). This yields the precise interlayer distance or thickness value at that location. The calculation results can not only be provided to the user numerically but can also be visualized intuitively using techniques like Color Mapping, rendered directly onto the model surface or generated as thematic maps. This clearly displays the spatial distribution patterns and variations of stratum thickness or interlayer distance, providing quantitative, visual data support for identifying potential weak zones, conducting engineering stability analysis, and assessing water hazard risks.

At the system implementation level, this research adopted the mature Browser/Server (B/S) architecture and designed a front-end/back-end separated system framework, enhancing system flexibility and maintainability. The front end utilized a modern Web technology stack (HTML5, CSS3, JavaScript) along with the core Three.js library to build the user interactive interface and the 3D rendering engine. Throughout the system development process, high priority was given to Web-end operational performance. A combination of various front-end performance optimization techniques was employed to address the challenges posed by large-scale geological data. Specific measures included: implementing asynchronous data loading mechanisms to avoid blocking the user interface response while loading large resources like models and textures; applying effective scene management strategies, such as dynamic rendering based on view distance and model importance (although not explicitly mentioned in the original text, ideas similar to LOD are common optimization directions); optimizing Three.js scene graph management, particularly focusing on the timely detection and explicit release of unused GPU resources like geometries, materials, and textures to mitigate memory leak risks and reduce GPU memory footprint, thereby lowering rendering load; exploring methods like model data compression for transmission to reduce network transfer time and bandwidth consumption. Through these optimization measures, the aim was to ensure that even when handling complex geological models containing millions or even more triangular faces, the system could maintain smooth operational speed and good interactive responsiveness on mainstream computer configurations and modern browsers.

To comprehensively validate the effectiveness and practicality of the theoretical methods, key technologies proposed, and the developed system platform, actual geological exploration data from a mine in Xingtai City, Hebei Province, China, were used as the research background and application case. Utilizing the detailed borehole data (including coordinates, elevation, stratum layering information, lithological descriptions), confirmed fault structure data, and planned or existing roadway engineering data from this mine area, a 3D geological model incorporating complex fault network structures, multiple sets of strata distribution, and underground roadway systems within the region was successfully constructed. Subsequently, within the developed WebGL-based prototype system, this model was displayed with high-quality visualization, and the various interactive analysis functions described earlier were provided. In particular, the core function of interlayer distance calculation was subjected to focused validation. By selecting multiple representative locations on the model for interlayer distance calculation, the results were compared in detail with the actual stratum thickness data revealed by some boreholes within the mine area. The comparison showed good consistency between the interlayer distances calculated by the ray intersection method used in this system and the measured borehole data, with errors falling within acceptable ranges, fully demonstrating the accuracy and reliability of this calculation method. The successful implementation of the entire application case not only showcased the capability of this research outcome in handling real-world complex geological data, constructing high-precision 3D geological models, and providing intuitive interactive visualization, but also verified the practical value of the integrated geological analysis functions (such as interlayer distance calculation). This indicates that the proposed WebGL-based Web-end 3D geological modeling and visualization solution is feasible and can provide crucial 3D spatial information support and powerful auxiliary decision-making tools for practical tasks such as mineral resource exploration and assessment, reserve estimation, mine engineering design, mining plan optimization, mine hydrogeological analysis, and water hazard risk assessment.

The research results demonstrate that the technical route based on WebGL and Three.js can effectively support the development of Web-based 3D geological applications, achieving efficient rendering and smooth interaction with geological models. Through the Constrained Delaunay Triangulation method, combined with refined handling of fault intersection relationships, fault and stratum models reflecting complex geological structures were successfully constructed; the borehole model based on segmented cylinders intuitively displays internal borehole information; the proposed adaptive UV coordinate calculation method combining bounding box and normal vector information improved the texture mapping effect of geological models to some extent; the implemented interactive and analytical functions, such as roadway roaming, attribute querying, and especially the interlayer distance calculation based on ray intersection, enhanced the practical value of the system. Concurrently, by adopting a series of performance optimization strategies including GPU-accelerated rendering, asynchronous loading, and memory optimization management, the performance bottleneck of loading and rendering large-scale geological data on the Web end was effectively addressed, ensuring the system's usability and user experience. The application case of the Xingtai mine validated the feasibility, accuracy, and stability of the research methods and system.

Although this research has made significant progress in Web-based 3D geological modeling and visualization, several areas warrant further exploration and improvement. Firstly, regarding model construction accuracy, the currently dominant triangular mesh (TIN) structure may still face challenges in ensuring mesh quality and maintaining topological consistency when representing extremely complex geological structures (such as intense folding, large-scale thrust nappe structures, dense fracture zones, and their complex intersections). Future research could consider introducing more advanced representation methods like Voxel, Implicit Surface, or Hybrid Modeling, and explore advanced meshing techniques such as Anisotropic Meshing, Adaptive Mesh Refinement, and Multi-resolution Models, aiming to optimize computational cost and storage requirements while ensuring model accuracy. Secondly, in handling ultra-large-scale geological datasets for real-time interaction and rendering, while existing GPU acceleration and optimization techniques have significantly improved performance, the current Web-end rendering capabilities might still encounter bottlenecks when facing future geological model data potentially reaching hundreds of millions or even higher polygon counts. Future work could closely follow and actively explore next-generation Web graphics APIs like WebGPU, leveraging their lower-level hardware access capabilities, better multi-threading support, and modern graphics features to further unlock Web-end rendering potential. Simultaneously, continuous research and optimization of techniques such as data streaming, View Frustum Culling, Occlusion Culling, fine-grained Level of Detail (LOD) strategies, and Server-Side Rendering combinations are needed to cope with ever-increasing data scales. Thirdly, concerning texture mapping, although the adaptive UV calculation method proposed in this study offers improvements, it may still suffer from projection distortion or UV coordinate discontinuities (jumps) when dealing with geometries exhibiting complex curvature changes or topological singularities (like sharp edges or vertices). Future research could explore more advanced and robust surface parameterization algorithms (such as conformal mapping, methods minimizing area or angle distortion), or attempt to combine machine learning techniques to automatically optimize UV layout. Meanwhile, a balance must be struck between algorithm complexity, computational efficiency, and final visual quality, possibly requiring finer model segmentation or feature recognition during preprocessing to assist UV generation. Furthermore, with the continuous evolution of Web technologies, 3D geological modeling platforms based on WebGL (and future WebGPU) are poised to further expand their functional boundaries and application scenarios. For instance, deep integration with Virtual Reality (VR) and Augmented Reality (AR) technologies could provide users with more immersive and interactive experiences for browsing, analyzing, and collaborating on geological models, allowing geological experts to explore the subsurface world "in situ." Integration with Artificial Intelligence (AI) technologies could also be strengthened, exploring advanced applications like intelligent interpretation of geological models, automatic identification of geological anomalies, and intelligent analysis and prediction based on geological knowledge. Finally, efforts should be made to promote deeper data integration and business linkage between such platforms and existing mine Web information systems (e.g., monitoring, production management, safety warning). By fusing 3D geological models with real-time production data, sensor monitoring data, etc., for combined visualization and correlation analysis, a true mine Digital Twin can be constructed, thereby providing more comprehensive, dynamic, and intelligent support for intelligent mine operations and refined management.

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